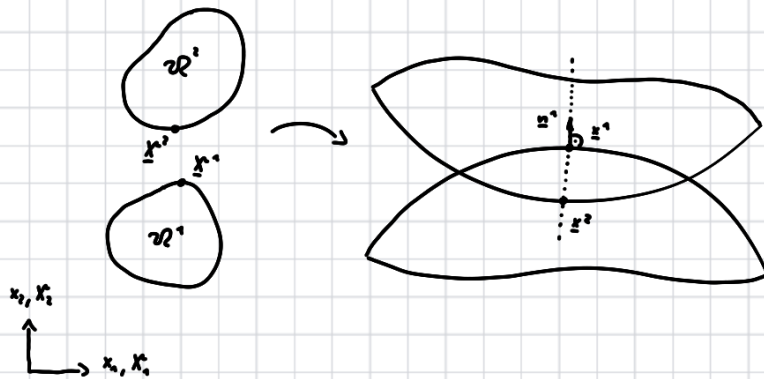


Contact Hertz



$$\underline{n}^1 \underline{x}^2 - \underline{n}^1 \underline{x}^1 = g_N \quad g_N < 0 \leftarrow \text{Penetration}$$

$$(\underline{x}^2 - \underline{x}^1) \underline{n}^1 = g_N$$

$$g_N \geq 0$$

$$(\underline{x}^2 - \underline{x}^1) \underline{n}^1 \geq 0$$

$$\underline{x}^1 = \underline{u}^1 + \underline{x}^1$$

$$\underline{x}^2 = \underline{u}^2 + \underline{x}^2$$

Signori condition

$$g_N \geq 0 \quad p_N \geq 0 \quad p_N g_N = 0$$

$$\delta W_C = \int_{\Gamma} p_N \delta g_N d\Gamma$$

$$\delta g_N = (\delta \underline{x}^2 - \delta \underline{x}^1) \underline{n}^1$$

Small deformation

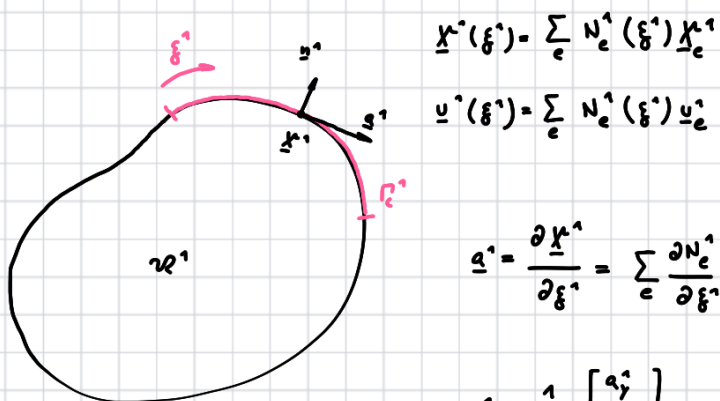
$$(\underline{u}^2 - \underline{u}^1) \underline{n}^1 + g_N = g_N \geq 0$$

$$g_N = (\underline{x}^2 - \underline{x}^1) \underline{n}^1$$

Penalty - Method

$$p_N = \begin{cases} 0 & g_N > 0 \\ \epsilon (-g_N) & g_N < 0 \end{cases}$$

$$\delta W_c = - \int_{\Gamma_c} p_N \delta g_N d\Gamma$$

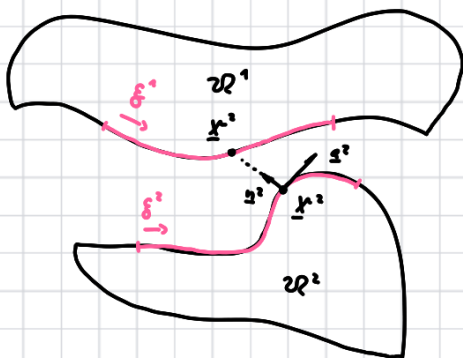


$$\underline{x}^1(\xi^1) = \sum_e N_e^1(\xi^1) \underline{x}_e^1$$

$$\underline{u}^1(\xi^1) = \sum_e N_e^1(\xi^1) \underline{u}_e^1$$

$$\underline{a}^1 = \frac{\partial \underline{x}^1}{\partial \xi^1} = \sum_e \frac{\partial N_e^1}{\partial \xi^1} \underline{x}_e^1$$

$$\underline{n}^1 = \frac{1}{\|\underline{a}^1\|} \begin{bmatrix} a_1^1 \\ a_2^1 \end{bmatrix}$$



→ search for ξ^2

$$(\underline{x}^2(\xi^2) - \underline{x}^1) \cdot \underline{a}^2 = 0$$

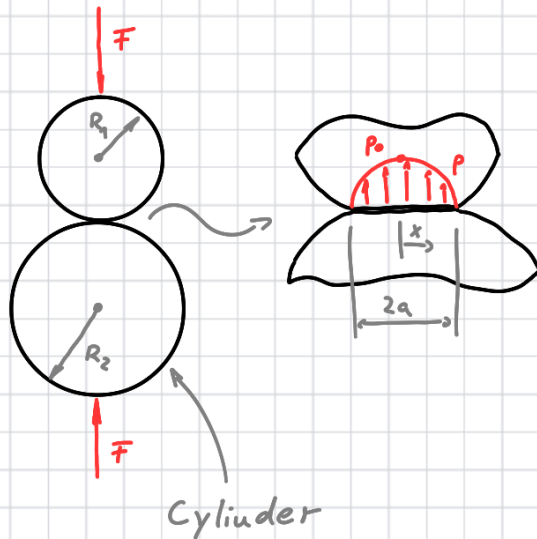
$$\underline{a}^2 = \frac{\partial \underline{x}^2}{\partial \xi^2} = \sum_e \frac{\partial N_e^2}{\partial \xi^2} \underline{x}_e^2$$

$$\underline{x}^2(\xi^2) = \sum_e N_e^2 \underline{x}_e^2$$

$$\underline{u}^2(\xi^2) = \sum_e N_e^2 \underline{u}_e^2$$

$$\delta g_N = (\delta \underline{u}^2 - \delta \underline{u}^1) \cdot \underline{n}^1 = \left(\sum_e N_e^2 \delta \underline{u}_e^2 - \sum_e N_e^1 \delta \underline{u}_e^1 \right) \cdot \underline{n}^1$$

Contact pressure



$$\frac{1}{R^*} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

$$p = p_0 \sqrt{1 - \frac{x^2}{a^2}} \quad a = \sqrt{\frac{4FR^*}{\pi E^*}}$$

$$p_0 = \frac{2F}{\pi a}$$

Plain strain

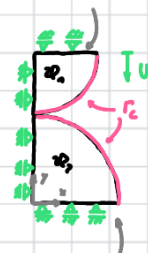
$$\mu = \frac{E}{2(1+\nu)} \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$R_1 = 12 \text{ mm} \quad E_1 = 2,1 \cdot 10^5 \text{ N/mm}^2 \quad \nu_1 = 0,3$$

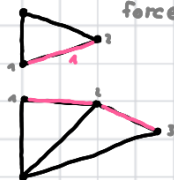
$$R_2 = 15 \text{ mm} \quad E_2 = 1,0 \cdot 10^5 \text{ N/mm}^2 \quad \nu_2 = 0,25$$

FEM

reaction force



Reaction force



→ search segment
to set Gauss nodes



$$N_1 = 1 - \xi \quad \frac{\partial N_1}{\partial \xi} = -1$$

$$N_2 = \xi \quad \frac{\partial N_2}{\partial \xi} = 1$$

$$\underline{x}^* = N_1^*(\xi^*) \underline{x}_1^* + N_2^*(\xi^*) \underline{x}_2^*$$

$$\underline{u}^* = N_1^*(\xi^*) \underline{u}_1^* + N_2^*(\xi^*) \underline{u}_2^*$$

$$\underline{a}^* = \frac{\partial \underline{x}^*}{\partial \xi^*}(\xi^*) = \frac{\partial N_1^*}{\partial \xi^*}(\xi^*) \underline{x}_1^* + \frac{\partial N_2^*}{\partial \xi^*}(\xi^*) \underline{x}_2^*$$

$$\underline{n}^* = \frac{1}{\|\underline{a}^*\|} \begin{bmatrix} a_y^* \\ -a_x^* \end{bmatrix}$$

$$\overset{\text{search } \xi^2}{(\underline{x}^1(\xi^1) - \underline{x}^*) \cdot \underline{a}^2 = 0}$$

$$(\underline{v}^1 - \underline{v}^*) \cdot \underline{n}^1 + g_N = g_N$$

$$g_N = (\underline{x}^2 - \underline{x}^*) \cdot \underline{n}^1$$

$$p_N = \begin{cases} 0 & g_N > 0 \\ \in (-g_N) & g_N < 0 \end{cases}$$

$$\delta g_N = (\delta v_2 - \delta v_1) \cdot \underline{n}_1 = \left(\sum_e N_e^2 \delta v_e^2 - \sum_e N_e^1 \delta v_e^1 \right) \cdot \underline{n}^1$$

$$\int_{\mathbb{R}} \underline{\sigma}(\underline{v}) : \underline{\varepsilon}(\underline{v}) \, dx - \int_{\Gamma} p_N \delta g_N \, d\Gamma = 0$$

$$\xi^2 = (\underline{x}^1 - \underline{x}_1^2) \cdot \underline{a}^2 / (\underline{a}^2 \cdot \underline{a}^2)$$

$$\underline{a}_2 = \underline{x}_2^2 - \underline{x}_1^2$$

$$\int_{\mathcal{V}} \underline{\sigma}(\underline{v}) : \underline{\varepsilon}(\underline{v}) \, dx - \int_{\Gamma_c} p_N \delta g_N \, d\Gamma = 0$$

$$\int_{\mathcal{V}} \underline{\sigma}(\underline{v}) : \underline{\varepsilon}(\underline{v}) \, dx = \underline{v}^T \underline{K} \underline{v}$$

$$- \int_{\Gamma_c} p_N \delta g_N \, d\Gamma = - \int_{\Gamma_c} \epsilon (-g_N) \delta g_N \, d\Gamma$$

$$g_N = (\underline{v}^2 - \underline{v}^1)^T \underline{n}^1 + g_X$$

$$\delta g_N = (\delta \underline{v}^2 - \delta \underline{v}^1)^T \underline{n}^1$$

$$= \epsilon \int_{\Gamma_c} g_N \delta g_N \, d\Gamma$$

$$g_N = \underline{n}^{1T} (\underline{N}^2 - \underline{N}^1) \underline{v} + g_X$$

$$g_N = \underline{\varepsilon}^T \underline{v} + g_X$$

$$\delta g_N = \underline{n}^{1T} (\underline{N}^2 - \underline{N}^1) \underline{v}$$

$$\delta g_N = \underline{\varepsilon}^T \underline{v} = \underline{v}^T \underline{\varepsilon}$$

$$\underline{v}^1 = N_1^1 \underline{u}_1^1 + N_2^1 \underline{u}_2^1$$

$$\underline{v}^1 = \begin{bmatrix} N_1^1 & 0 & N_2^1 & 0 & \dots \\ 0 & N_1^1 & 0 & N_2^1 & \dots \end{bmatrix} \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ \vdots \end{bmatrix}$$

$$\underline{v}^2 = \begin{bmatrix} \dots & N_1^2 & 0 & N_2^2 & 0 \\ 0 & N_1^2 & 0 & N_2^2 & \dots \end{bmatrix} \begin{bmatrix} u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \\ \vdots \end{bmatrix}$$

$$= \epsilon \underline{v}^T \int_{\Gamma_c} [\underline{\varepsilon} \underline{\varepsilon}^T \underline{v} + \underline{\varepsilon} g_X] \, d\Gamma \quad \underline{\varepsilon} = (\underline{N}^2 - \underline{N}^1)^T \underline{n}^1$$

$$= \underline{v}^T \underline{K}_c \underline{v} + \underline{v}^T \int_{\Gamma_c} \epsilon \underline{\varepsilon} g_X \, d\Gamma$$

$$\underline{v}^1 = \underline{N}^1 \underline{v} \quad \delta \underline{v}^1 = \underline{N}^1 \underline{v}$$

$$\underline{v}^2 = \underline{N}^2 \underline{v} \quad \delta \underline{v}^2 = \underline{N}^2 \underline{v}$$

$$\underline{K}_c = \epsilon \int_{\Gamma_c} \underline{\varepsilon} \underline{\varepsilon}^T \, d\Gamma = \epsilon \|a_n\| \sum (w_1 \underline{\varepsilon} \underline{\varepsilon}^T + w_2 \underline{\varepsilon} \underline{\varepsilon}^T)$$

$$d\Gamma = \|a_n\| \, d\xi$$

$$w_1 = 0,5 \quad \xi_1 = 1/2 + 1/2 \frac{1}{\sqrt{3}}$$

$$w_2 = 0,5 \quad \xi_2 = 1/2 - 1/2 \frac{1}{\sqrt{3}}$$

Newton: insert $\underline{v}_0$

$$\underline{v}^{n+1} = \underline{v}^n + \Delta \underline{v}$$

$$\underline{J}(\underline{v}^n) \Delta \underline{v} = -\underline{R}(\underline{v}^n)$$

$\Delta \underline{v}_i = 0$ where $\underline{v}_0$ is set

$$\underline{J} = \underline{K} + \underline{K}_c$$

Only active if $g_N = \underline{\varepsilon}^T \underline{v} + g_X < 0$

$$\underline{R} = \underline{K} \underline{v}^n + \epsilon \int_{\Gamma_c} [\underline{\varepsilon} \underline{\varepsilon}^T \underline{v}^n + \underline{\varepsilon} g_X] \, d\Gamma$$